Chapter 6. Sequences and Series of Functions

Uniform convergence, differentiation and integration of limits, double series, power series, trigonometric series

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6.1 Uniform convergence

Motivation. Functions are often defined as limits: power series, Fourier series, solutions of differential equations. Pointwise limits can destroy continuity (x^n on $[0,1]$), so we need a stronger notion. Uniform convergence is the one that preserves continuity, and the Weierstrass M-test is the usual way to verify it.

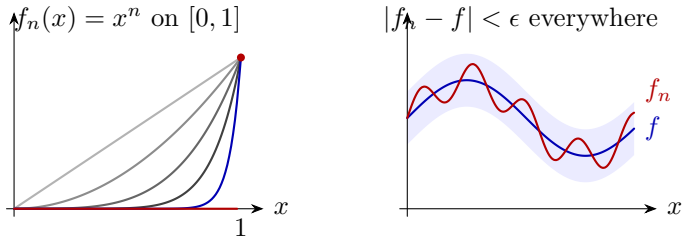


Figure. *Left: x^n converges pointwise on $[0,1]$ to a discontinuous limit (0 for $x<1$, 1 at $x=1$), so the convergence cannot be uniform. Right: uniform convergence means that for large n the whole graph of f_n lies in the ϵ-band around f.*

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Motivation. When can we exchange a limit with an integral or a derivative? Uniform convergence allows the first; the second needs uniform convergence of the derivatives. These are the theorems that justify term-by-term manipulation of series.

Theorem 6.4 (6.2.1)

Let $f_n : [a, b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a, b]$;
(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t| \, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$ this shows that (f_n) is uniformly Cauchy, hence converges uniformly to some f. Fix x and set $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$, $\phi(t) = \frac{f(t) - f(x)}{t - x}$ for $t \neq x$. By $(*)$, (ϕ_n) is uniformly Cauchy on $[a, b] - \{x\}$, so $\phi_n \to \phi$ uniformly there. Let $g = \lim f_n'$. Given ϵ, choose N with $|\phi_N - \phi| < \epsilon$ on $[a, b] - \{x\}$ and $|f_N'(x) - g(x)| < \epsilon$, and δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$ for such t, so $f'(x) = g(x)$. ∎

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a, b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b - a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b - a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b - a)$. ∎

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n \ge 0} (-1)^n t^{2n}$ (uniform on $[-r, r]$, $r < 1$) gives $\arctan x = \sum (-1)^n x^{2n+1}/(2n + 1)$ for $|x| < 1$; similarly $\log(1 + x) = \sum (-1)^{n+1} x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $n a_n \to 0$: $\frac{n}{2} a_n \le a_{\lceil n/2 \rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Motivation. Iterated limits and sums cannot always be exchanged. This section gives conditions, absolute convergence above all, under which the order of summation does not matter; they are

needed for products of series and for power series in two variables.

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn} - c_n| < \epsilon$ for $m \geq N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. Regard $f_m(n) = a_{mn}$ as functions on $X = \{1, 1/2, 1/3, \dots\} \cup \{0\}$ (with $n \leftrightarrow 1/n$), extended to 0 by $f_m(0) = b_m$; each f_m is continuous by (a), and by (c) and Theorem 6.2 the f_m converge uniformly on X (at 0 too, since $|b_m - b_k| \leq \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1), with value c_n at $1/n$ and $\lim b_m$ at 0; continuity at 0 gives $\lim_n c_n = \lim_m b_m$. ■

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \leq \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \leq \sum_m b_m$, so the inner sums converge absolutely. Let $S_{MN} = \sum_{m \leq M} \sum_{n \leq N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m \leq M} \sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n \leq N} \sum_m a_{mn}$, uniformly in N, since $|S_{MN} - \sum_{n \leq N} \sum_m a_{mn}| \leq \sum_{m > M} b_m$. Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \geq 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m \leq M} \sum_{n \leq N} a_{mn} \leq \sum_{n \leq N} \sum_m a_{mn} \leq \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\leq$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n) \in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k \in \mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n \to \infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \geq 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k \in \mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ■

6.4 Power series

Motivation. Power series are the most important example of uniform convergence. They converge uniformly on smaller discs, can be differentiated term by term, and therefore define smooth functions such as e^x, $\sin x$, $\cos x$ rigorously.

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup|a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup|a_n x^n|^{1/n} = \lambda|x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup|n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived series has the same radius and converges uniformly on $[-r, r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ∎

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R, R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!}(x-c)^n$ for $|x-c| < R - |c|$.

Proof. $f(x) = \sum_n a_n(c+(x-c))^n = \sum_n \sum_{k \le n} a_n \binom{n}{k} c^{n-k}(x-c)^k$. Since $\sum_n \sum_k |a_n| \binom{n}{k} |c|^{n-k}|x-c|^k = \sum_n |a_n|(|c| + |x-c|)^n < \infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x) = \sum_k (\sum_n \binom{n}{k} a_n c^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ∎

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n = \sum_{k=0}^n a_k b_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$, then $\sum c_n = AB$.

Proof. With partial sums A_n, B_n, C_n and $\beta_n = B_n - B$, $C_n = \sum_{k \le n} a_k B_{n-k} = A_n B + \gamma_n$, $\gamma_n = \sum_{k \le n} a_k \beta_{n-k}$. Let $\alpha = \sum |a_k|$ and $\epsilon > 0$; choose N with $|\beta_m| < \epsilon$ for $m \ge N$. Then $|\gamma_n| \le \epsilon\alpha + \sum_{n-N < k \le n} |a_k||\beta_{n-k}|$, and the last sum tends to 0 as $n \to \infty$ because $a_k \to 0$ and the β's are bounded. So $\gamma_n \to 0$ and $C_n \to AB$. ∎

Theorem 6.15 (6.4.4, Abel)

Let $f(x) = \sum a_n x^n$ on $(-R, R)$. If $\sum a_n R^n$ converges, then $\lim_{x \to R^-} f(x) = \sum a_n R^n$.

Proof. Replacing a_n by $a_n R^n$ we may take $R = 1$; subtracting a constant, assume $s = \sum a_n = 0$. With partial sums $s_n \to 0$, summation by parts gives $\sum_{n \le N} a_n x^n = (1-x)\sum_{n<N} s_n x^n + s_N x^N$, so $f(x) = (1-x)\sum s_n x^n$ for $|x| < 1$. Given ϵ, choose M with $|s_n| < \epsilon$ for $n \ge M$; then $|f(x)| \le (1-x)\sum_{n<M}|s_n| + \epsilon$ for $0 < x < 1$, which is $< 2\epsilon$ for x near 1. ∎

Corollary 6.16 (6.4.5)

If $\sum a_n = A$, $\sum b_n = B$ and their Cauchy product $\sum c_n = C$ all converge, then $C = AB$.

Proof. For $|x|<1$ the series $\sum a_n x^n$, $\sum b_n x^n$ converge absolutely, so $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$ by Proposition 6.14. Let $x \to 1^-$ and use Theorem 6.15 three times. ∎

Example 6.4

Abel's theorem gives $\log 2 = 1 - \tfrac12 + \tfrac13 - \cdots$ and $\tfrac{\pi}{4} = 1 - \tfrac13 + \tfrac15 - \cdots$ from Example 6.3. Defining $e^x = \sum x^n/n!$, Proposition 6.14 gives $e^{x+y} = e^x e^y$; with $\sin, \cos$ defined by their series, $(\sin)' = \cos$, $(\cos)' = -\sin$, and $\sin^2 + \cos^2 = 1$ follow by termwise differentiation.

6.5 Trigonometric series

Motivation. Fourier analysis represents a periodic function as a sum of sines and cosines. Here we prove the basic properties of trigonometric series that converge uniformly, as preparation for Chapter 9.

A complex vector space V with a map $\langle \cdot,\cdot\rangle$ that is linear in the first variable, satisfies $\langle y,x\rangle = \overline{\langle x,y\rangle}$, and $\langle x,x\rangle > 0$ for $x \neq 0$, is an **inner product space**; $\|x\| = \langle x,x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x,y\rangle| \le \|x\|\|y\|$ in a complex inner product space; hence $\|x+y\| \le \|x\| + \|y\|$.

Proof. If $y \neq 0$, put $\lambda = \langle x,y\rangle/\|y\|^2$; then $0 \le \|x - \lambda y\|^2 = \|x\|^2 - |\langle x,y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x,y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k\le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{†}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By (†), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ∎

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ∎

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx}-1}{e^{ix}-1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{‡}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi-\delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum |a_n - a_{n-1}| < \infty$, then for every $\delta \in (0, \pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi - \delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by (‡)) and constant $g_n = a_n$; take real and imaginary parts. ■

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) Given ϵ, uniform convergence gives $|\sum_{n/2 \le k \le n} a_k \sin kx| < \epsilon$ for all x and all large n. Take $x = \pi/(2n)$: for $n/2 \le k \le n$, $kx \in [\pi/4, \pi/2]$, so $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$. There are at least $n/2$ such k, so $\frac{n}{2} a_n/\sqrt{2} < \epsilon$. Hence $na_n \to 0$.

($\Leftarrow$) Since a_k decreases, (†) with $f_k = \sin kx$ and (‡) give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m / \sin(x/2) \le 2\pi a_m / x$ for $0 < x \le \pi$ (using $\sin(x/2) \ge x/\pi$). Let $\epsilon_m = \sup_{k \ge m} k a_k \to 0$. By oddness and periodicity it suffices to bound $R_m(x) = \sum_{k \ge m} a_k \sin kx$ for $x \in (0, \pi]$; let N be the largest integer $\le \pi/x$. If $m > N$, then $m \ge N + 1 > \pi/x$, so $|R_m(x)| \le 2\pi a_m / x < 2m a_m \le 2\epsilon_m$. If $m \le N$, split $R_m = \sum_{k=m}^{N} + \sum_{k>N}$. In the first sum $|a_k \sin kx| \le k a_k x \le \epsilon_m x$, and there are at most $N \le \pi/x$ terms, so it is at most $\pi \epsilon_m$; the second is at most $2\pi a_{N+1}/x < 2(N+1) a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi + 2)\epsilon_m \to 0$. ■

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \le g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \le M$. Then $\sum f_n g_n$ converges uniformly.

Proof. Apply (†) to the tails: with $R_k = \sum_{j=m}^{k} f_j$ ($R_{m-1} = 0$), $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k (g_k - g_{k+1})$. Given ϵ, for m large $|R_k(x)| < \epsilon$ for all $k \ge m$, x, by (a). Since $g_k - g_{k+1} \ge 0$, the sum is at most $\epsilon M + \epsilon(g_m - g_n) \le 3M\epsilon$ in absolute value. ■